Literature Review

1. Feedback & Evaluation

Strong Points

- **Exceptional Architectural Cohesion Across the Medallion Layers**: The progression from raw multi-source ingestion (Local CSV batch + Cloud Supabase REST API) to an immutable Bronze layer, an automated Silver layer, and a dedicated Gold Star Schema is modeled with genuine enterprise discipline. Most student projects jump straight from flat CSVs into Power BI without demonstrating data lineage or staging controls.
- **Realistic Data Quality Engineering**: Rather than creating synthetic datasets with trivial missing values, your 20-rule anomaly catalog addresses authentic operational problems—including webhook retry duplicates, non-standard ISO country strings, mixed date parsing, broken P&L identities, negative CAC signs, and orphan foreign keys.
- **Separation of Concerns in Error Handling**: The architectural split—handling network timeouts, API rate limits, and JSON pagination in Python try...except blocks, while handling multi-table transactional rollbacks inside SQL Stored Procedure TRY...CATCH blocks—reflects real-world data engineering standards.
- **Clear Role-Specific Decision Framework**: Structuring the executive deliverables around concrete decisions (CEO Capital Allocation, COO Supplier Defect Auditing & Backlog Recovery, CFO Promotional Margin Caps, and Retention Win-Back Execution) ensures the visualizations serve strategic actions rather than passive data display.

Critical Examiner Observations

- **Composite Key Discipline in Fact Tables**: Your line-item table relies on a composite business key (order_id + product_id). In high-scale warehouse modeling, examiners prefer an explicit surrogate key (order_item_key via IDENTITY or HASHBYTES) to ensure 1-to-many relationship performance in Power BI.
- **In-Memory vs. Database-Driven Imputation**: While your SQL scripts use window functions (AVG(...) OVER (PARTITION BY customer_segment)) cleanly, examiners will check whether your Python splitting script unintentionally normalized values before Bronze staging. Ensuring raw strings land completely uncleaned in Bronze is critical to proving your Silver cleaning logic.
- **DAX Measure Specificity**: Examiners will verify that your Power BI report uses strict explicit measures (DIVIDE()) rather than dragging raw columns into visual cards, especially for calculated ratios like profit margins, return rates, and CLV-to-CAC multiples.

2. Suggested Improvements

Domain	Current Implementation	Suggested Technical Enhancement	Value to Project & Defense
Data Warehouse Optimization	Natural composite keys (order_id + product_id) on line items.	Add an integer surrogate key fact_order_line_item_key = CONCAT(order_id)	product_id and apply clustered columnstore indexing in SQL Server.
Incremental Load Automation	Stored procedures using truncate-and-reload or full rebuilds.	Implement a formal Watermark Table (etl_watermarks) storing last_successful_ingestion_ts and execute delta MERGE INTO statements for updated orders.	Demonstrates production ETL scalability for millions of daily transactions.
Predictive Churn Scoring	Rule-based recency and SLA delay gap threshold classification (Active, At-Risk, Churned).	Train a lightweight Python ML classification model (Logistic Regression or Random Forest) to output a 30-day churn probability score ($P(\text{Churn}_{i})$) stored in dim_customers.	Bridges traditional BI reporting with forward-looking predictive data science.
Power BI Row-Level Security (RLS)	All 4 stakeholder pages are accessible to any viewer.	Configure dynamic Row-Level Security in Power BI Desktop based on USERPRINCIPALNAME() so regional managers see only their territory and operational leads cannot view sensitive executive margin waterfalls.	Demonstrates enterprise governance, role-based access control, and data privacy compliance.

3. Final Grading Criteria

Assessment Pillar	Marks Allocated	Evaluation Criteria & What the Grading Committee Checks
1. Documentation & Architecture	**25 Marks**	• **Data Quality Catalog (10 Marks)**: Thoroughness of the 20-rule anomaly catalog with clear before-and-after SQL remediation logic. • **Architecture & Schema Diagrams (10 Marks)**: End-to-end data lineage diagrams from hybrid sources (CSV + API) to the Gold Star Schema. • **Requirements & Project Trace (5 Marks)**: Alignment between stakeholder business questions, user stories, and technical delivery.
2. Implementation & Engineering	**35 Marks**	• **Staging & Bronze Ingestion (8 Marks)**: Working staging setup, lineage metadata tracking, and Supabase REST API pagination logic. • **Silver Layer SQL Transformations (12 Marks)**: Correct use of CTEs, window functions, regex cleaning, deduplication, and transactional rollbacks. • **Gold Star Schema & SQL Views (8 Marks)**: Clean separation of facts and dimensions, surrogate key modeling, and orphan SKU

Assessment Pillar	Marks Allocated	Evaluation Criteria & What the Grading Committee Checks
		fallback handling. • **Python Automation Pipeline (7 Marks)**: Working 1-click execution script with robust try...except logging and database connectors.
3. Testing, Validation & Modeling	**20 Marks**	• **Data Accuracy & Reconciliation (8 Marks)**: Zero variance between Power BI visual aggregates and underlying SQL database totals. • **Power BI Relationship Integrity (6 Marks)**: Enforcing strict 1:* single-direction relationships with zero bi-directional cross-filtering or inactive join traps. • **DAX Mathematical Rigor (6 Marks)**: Proper use of explicit ratio measures (DIVIDE()), zero-division guards, and time-intelligence calculations.
4. Presentation, UI/UX & Defense	**20 Marks**	• **Cognitive Layout & 16:9 UX Design (8 Marks)**: Scannable executive layout, cohesive 4-color semantic palette, and strictly one visual per question. • **Interactive Storytelling (6 Marks)**: Seamless live walkthrough demonstrating slicers, drill-downs, and policy simulator impact. • **Technical Defense & Question Handling (6 Marks)**: Ability of team members to justify architectural trade-offs, TRY...CATCH

Assessment Pillar	Marks Allocated	Evaluation Criteria & What the Grading Committee Checks
		boundaries, and data design choices.
Total Score	**100 Marks**	**Comprehensive Academic Assessment Rubric for Final Delivery**